

CODEx ROVER BEAN

The Robot Body Builder's Book

Print it. Screw it together. Meet your robot.
Everything you need is in this book and one Bambu Studio file.

AGES 10+

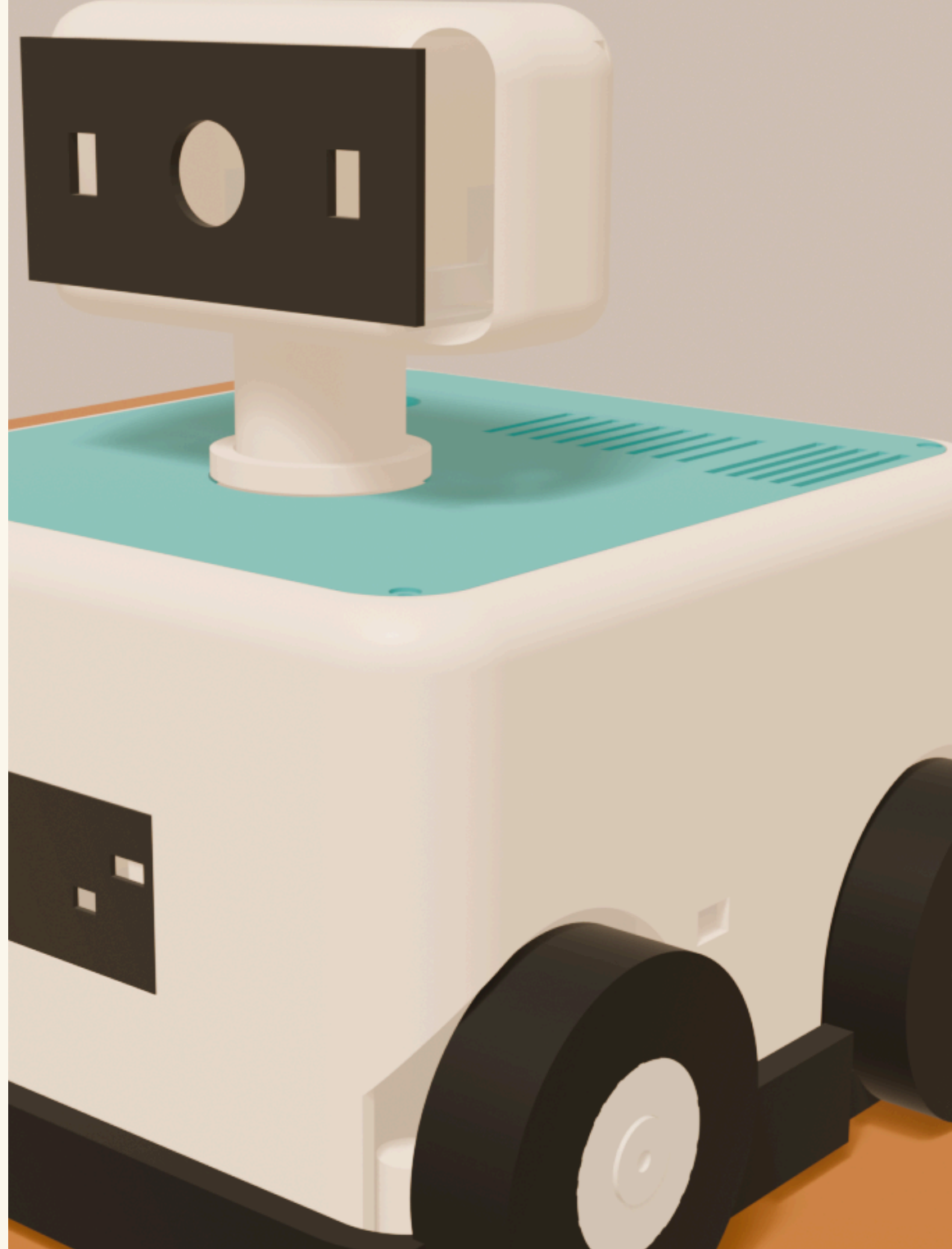
44 PIECES TO PRINT

11 PLATES

1 HEX KEY

GROWN-UP NEEDED: SOLDERING & WIRING

Chapters: 1 Shop - 2 Print - 3 Build - 4 Wire (grown-up) - 5 Check & play



How this book works

The pictures do the talking

Every build step shows the robot so far. On most pictures the NEW parts float just off their spot (early steps show them already home). The little dark tag with the arrow points toward the robot's FACE. Match the picture, then read the numbered lines if you want words too.

The parts strip

The tan strip at the top of each step shows exactly which parts and how many screws you need **before you start**. Lay them out like a cooking show.

The green CHECK bar

Do the little test at the bottom of each step before moving on. If the check fails, fix it now - later steps cover things up.

Red circles mean grown-up

Steps with a **red number** use the soldering iron or touch wires. A grown-up does those; you can watch and help.

One screw. One key.

Every screw in this robot is the same M3 x 8 screw, and one 2.5 mm hex key turns them all. "Snug" means: stop when it stops, then an eighth of a turn. Plastic hates gorillas.

Front and back

The FRONT is where the face panel and head look. The BACK has the charging plug, the mute switch, and the motor wheels. Left and right are the robot's left and right, not yours.

Chapter 1 - Go shopping: plastic, screws, tools

Filament (for the printer)

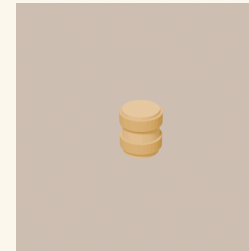
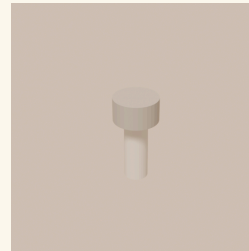
Filament	How much	What it becomes
Cream PETG	1x 1 kg spool (uses ~600 g)	The body, tray, head, and most parts.
Teal PETG	1x small spool or 250 g (uses ~80 g)	The top lid.
Charcoal / black PETG	leftovers are fine (~20 g)	Face panel, back panel, head face.
Translucent lime PETG	smallest spool available (uses ~5 g)	The glowing eye and light bars.
Charcoal TPU 95A	1x 500 g spool (uses ~280 g)	Soft tires, bumpers, battery pad.

The only two fastener packs

Fastener	Buy	Why
M3 x 8 mm socket head screws	1x 100-pack	The ONLY screw in the robot (41 used + spares).
M3 x 5.7 mm brass heat-set inserts (4.6 mm OD)	1x 100-pack	The only insert (41 used + spares).

Tools

Tool	Job
2.5 mm hex key	Turns every screw in this robot. Seriously, all of them.
Soldering iron (grown-up)	Melts the brass inserts into the plastic.
Small flush cutters / scissors	Trims zip ties and TPU strings.
Painter's tape + marker	Label wires as you go.
Multimeter (grown-up)	Checks every circuit before the battery ever goes in.



This screw and this brass insert are the only fasteners you'll buy for the whole robot. When a step says "2 screws," it always means these.

Chapter 1 - Go shopping: the electronics box (1 of 2)

Every item is a normal buy-one online part in the US. A grown-up orders these. Fuse sizes and wire gauges are right here in the list; the wiring chapter shows where each goes.

Part	Qty	What it does
Raspberry Pi 5 (8 GB)	1	The robot's computer.
32 GB+ A2 microSD card	1	The computer's memory card - the robot's programs live here.
Raspberry Pi Camera Module 3 Wide + long FPC cable	1	The robot's eye.
Raspberry Pi Pico 2 (no headers, no WiFi)	1	The safety helper: reflexes and watchdog.
Cytron MDDS10 motor driver	1	The purple board that powers the wheels.
Pololu #4867 gearmotor (99:1, 25D, 12 V, encoder)	2	The wheel motors.
Hitec D85MG servo	2	The neck motors (look left/right, up/down).
Bioenno BLF-1203AB 12 V 3 Ah LiFePO4 battery	1	The robot's power pack.
Bioenno BPC-1502DC charger	1	The matching charger. Only ever use this one.
Switchcraft EN2P3M20 inlet + EN2C3F20G2 plug	1 pair	The keyed charging plug on the back.
Pololu D24V90F5 regulator (5 V)	1	Makes clean 5 V for the Pi.
Pololu D36V50F6 regulator (6 V)	1	Makes 6 V for the neck servos.

Chapter 1 - Go shopping: the electronics box (2 of 2)

The rest of the electronics box:

Part	Qty	What it does
Panasonic CB1A-R-M-12V relay	1	The motor power switch the red button controls.
Blue Sea Systems 5045 fuse block	1	Splits power safely into four fused branches.
ATO fuses: 2 A, 3 A, 5 A, 7.5 A assortment	1 kit	Branches start at 2-3 A (5 A max); feeder 7.5 A; motor branch 5 A. Grown-up confirms with the meter.
IDEC XW1E-BV402M-R emergency stop	1	THE BIG RED BUTTON.
E-Switch PVB3F230SS311 mute switch	1	The microphone privacy switch (glows red when muted).
Omron D2HW-C202MR bumper switches	6	Feelers inside the bumpers.
VL53L1X time-of-flight boards (Adafruit 3967)	4	Distance eyes: two in front, one each side.
Adafruit 5975 NeoPixel breakouts + JST-SH cables	4	The glowing eyes and status lights.
ReSpeaker USB mic array	1	The robot's ears.
Enclosed 3 W 4 ohm speakers + 2x Adafruit MAX98357A amps	1 set	The robot's voice.
18 AWG (power) + 22 AWG (signal) silicone wire, JST/spade connectors, ferrules, zip ties, heat-shrink	1 kit	Grown-up wiring supplies.

Chapter 2 - Print it with Bambu Studio

Open the project

Open **codex_robot_body_v2_p1s.3mf** in Bambu Studio. All 11 plates are already laid out for the P1S with a 0.4 mm nozzle and Textured PEI plate.

Load the right color

Each plate's name says the filament to load (cream, teal, charcoal, lime, or TPU). Print plates one at a time and change filament between groups.

No supports. Ever.

Every part is designed to print with zero supports. If the slicer asks for supports, something is wrong - don't add them, re-check the plate.

TPU is slow and squishy

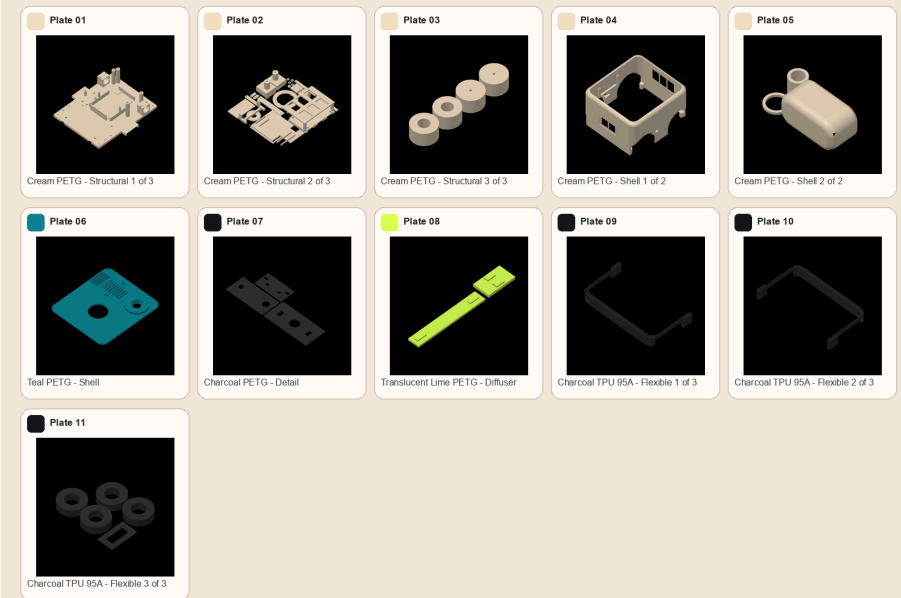
Print the tire and bumper plates slowly (the profile already does this). Dry TPU prints much better.

Big flat parts stay put

The tray, shell, and bumper plates fill the whole bed. Clean the plate with dish soap first so they stick.

Codex Robot Body v2 - P1S Plate Layout

44 parts / 11 plates / P1S 0.4 mm / Textured PEI / one material and color per plate



All 11 plates, exactly as they open in Bambu Studio.

Chapter 2 - The 11 plates, in printing order

#	Plate (load this filament)	Parts	What's on it
1	 Cream PETG - Structural 1 of 3	1	tray
2	 Cream PETG - Structural 2 of 3	22	deck, controller tower, battery clamp, speaker clamp #1, speaker clamp #2, yoke, mic cradle, head pan plate, pico clamp, front pod left, front pod right, motor cap #1, motor cap #2, tof clamp #1, tof clamp #2, tilt bushing, printed washer #1, printed washer #2, printed washer #3, printed washer #4, printed washer #5, printed washer #6 (<i>the robot uses 2 washers - the rest are spares</i>)
3	 Cream PETG - Structural 3 of 3	4	front wheel #1, front wheel #2, rear wheel #1, rear wheel #2
4	 Cream PETG - Shell 1 of 2	1	shell
5	 Cream PETG - Shell 2 of 2	3	head shell, bayonet collar, neck
6	 Teal PETG - Shell	1	lid
7	 Charcoal PETG - Detail	3	head faceplate, rear panel, face panel
8	 Translucent Lime PETG - Diffuser	2	status diffuser bar, eye diffuser bar
9	 Charcoal TPU 95A - Flexible 1 of 3	1	bumper front
10	 Charcoal TPU 95A - Flexible 2 of 3	1	bumper rear
11	 Charcoal TPU 95A - Flexible 3 of 3	5	battery pad frame, tire #1, tire #2, tire #3, tire #4

Tip: print plates 1-5 (cream) back to back, then change color once per group. Keep every part in a labeled box - the next chapter uses them in order.

Chapter 2 - Print the little test parts FIRST

Before the big plates, ask your grown-up to print the little TEST PARTS that come with the project (the **coupons** folder next to the Bambu file). They make sure your printer's holes, snaps, and fits are dialed in - like tasting the batter before baking the whole cake. Each one has a simple pass test:

Test part	Filament	What it proves
insert m3	PETG	Grown-up melts one insert into each of the three holes. The one that sits flush without squeezing out goo is your printer's perfect fit.
joint flange	PETG	Screw the small plate onto the block with one screw. It should pull down tight with no wobble and no cracking.
joint boss	PETG	The partner block for the plate test above.
dbore torque	PETG	Grown-up pushes this onto a motor shaft and twists HARD, twice as hard as driving ever will. If it never slips, wheels are safe.
axle stub	PETG	Spin the ring on the peg a few hundred times with some pressure. It should stay smooth, not sloppy.
bushing ring	PETG	The spinning ring for the peg test above.
snap pair	PETG	Click the two pieces together and apart ten times. The little hook must survive and still hold on.
bayonet pair	PETG	Twist the ring onto the stub a quarter turn - it should lock with a nice click and not pull straight off.
switch pocket	PETG	Grown-up seats one bumper switch in the pocket: it must click when pressed and spring back, never jammed.
tire fit	TPU+PETG	Stretch the mini tire onto the mini wheel. Snug and even = your big tires will fit too.
pcb clamp	PETG	Rest the little green safety board on the pegs and screw the bar over it. Held gently, nothing bending.

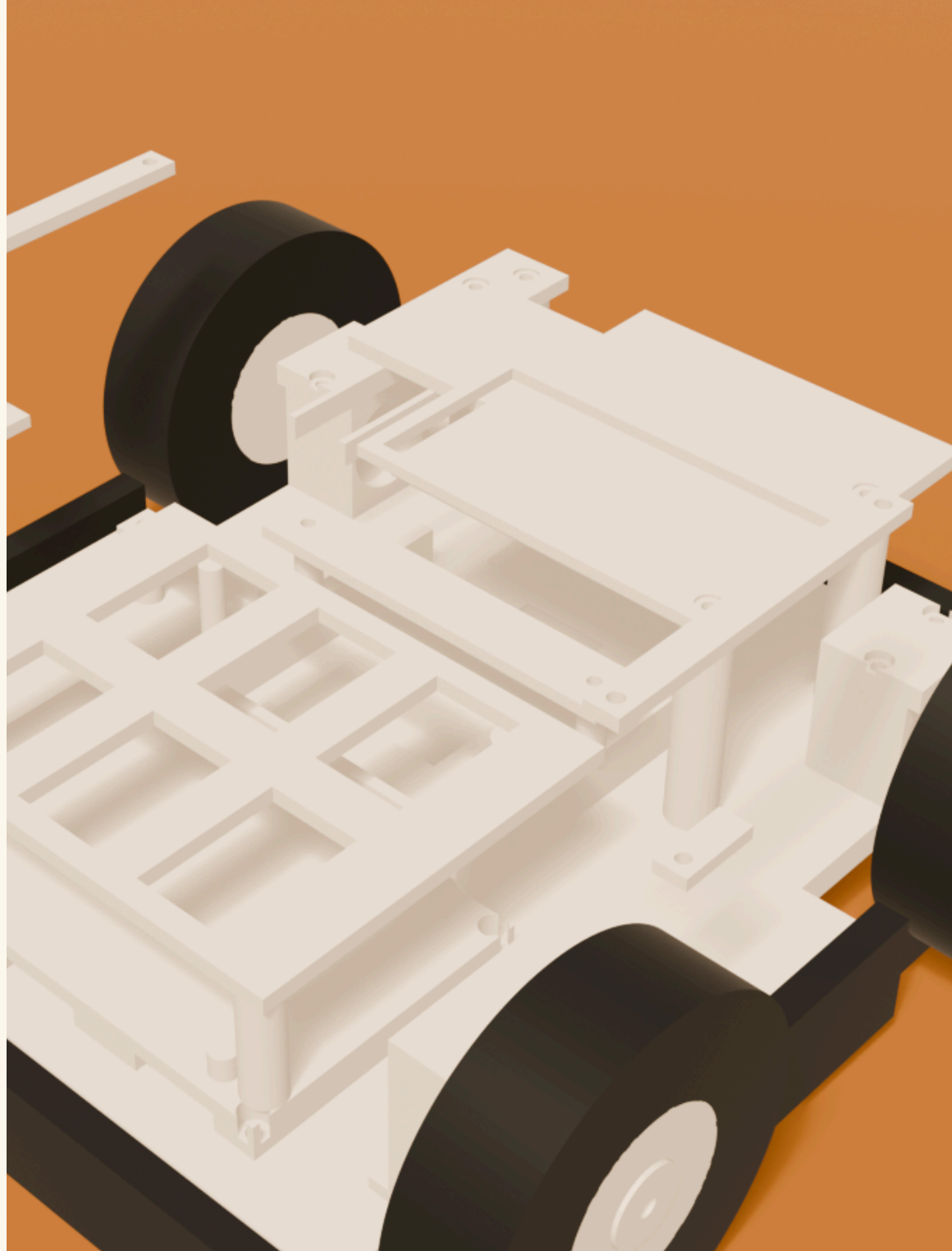
Chapter 3

Build it

Twenty steps. Lay out the parts from each step's tan strip before you start, match the big picture, then run the green CHECK.

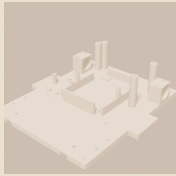
The little **FRONT** tag on every picture points at the robot's face, so left and right never get confusing.

Red-number steps need a grown-up.

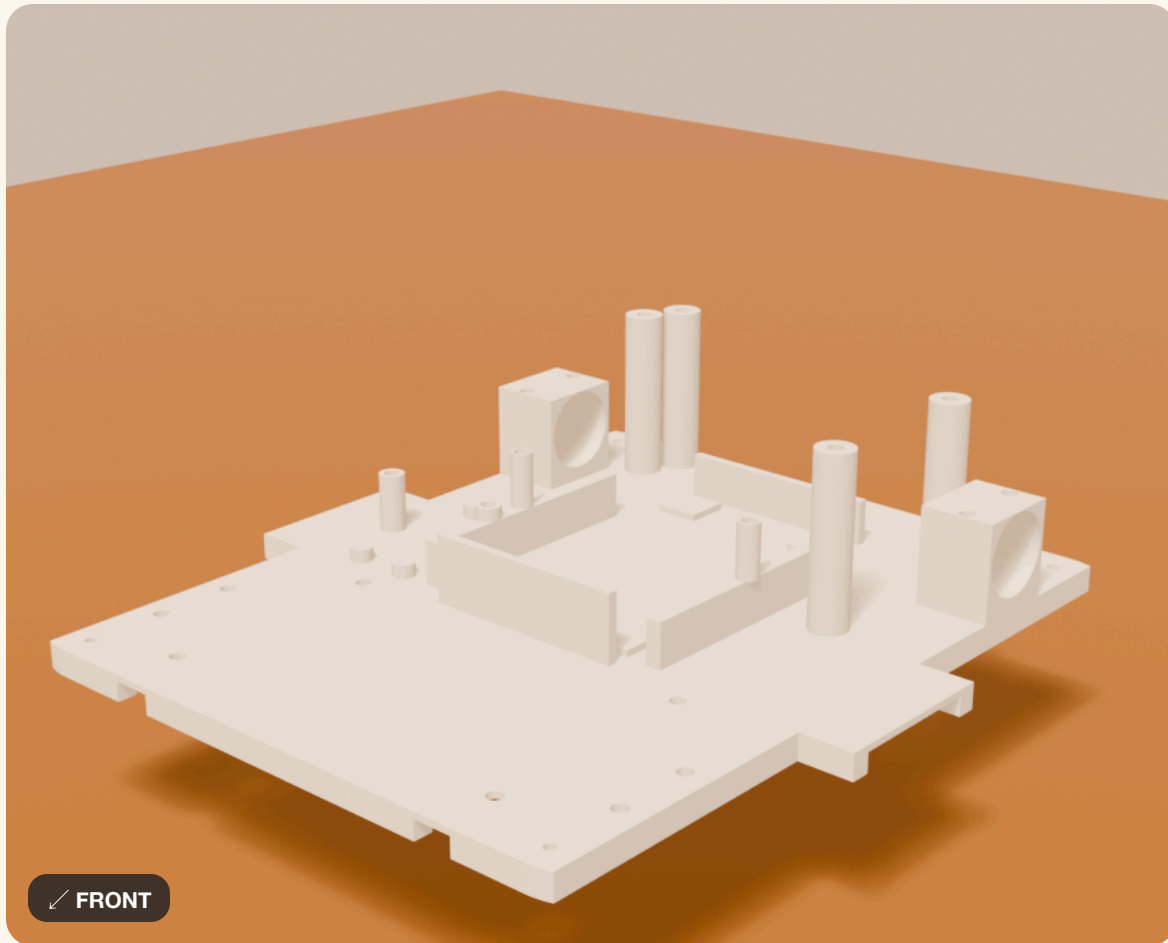


1

Start with the floor



x1



1. Clear a big table. Put the tray down flat, wheel cutouts toward you.
2. Find the two round motor cradles at the back and the tall round towers - that's the back of the robot.

CHECK The tray sits flat and doesn't rock.

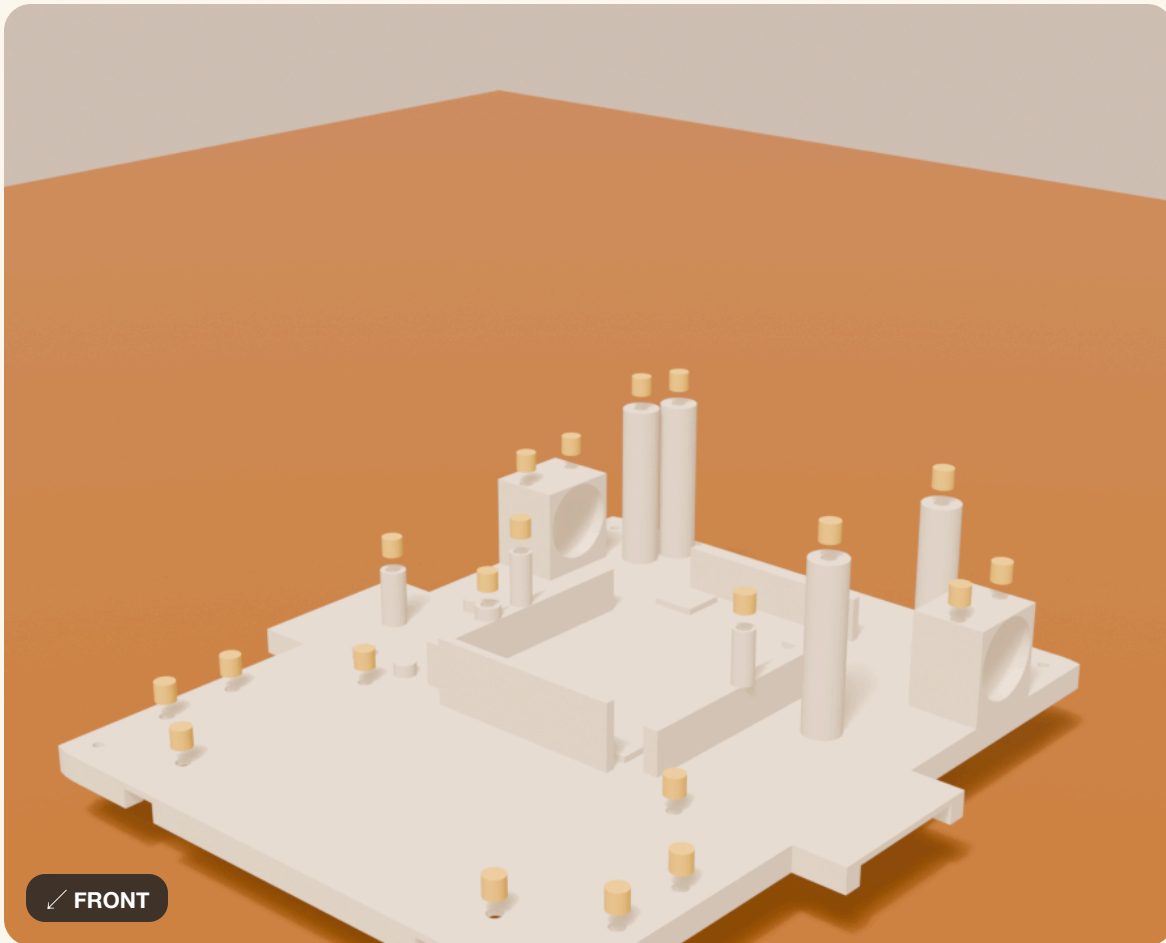
2

Melt in the tray's brass inserts

GROWN-UP STEP



x20

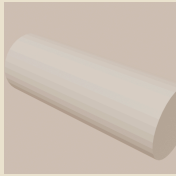


1. GROWN-UP: set the soldering iron to about 220 C.
2. Rest a brass insert in each gold-marked hole, then press it straight down with the hot iron tip until it sits flush.
3. 20 go into the tray now - the picture marks every spot. The other 21 come later: 2 in the wheels, 2 in the front pods, 14 in the shell, 2 in the lid, and 1 in the head. This book asks the grown-up back each time.

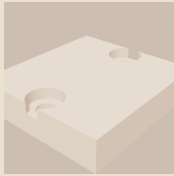
CHECK Every insert is flush and straight, none tilted.

3

Drop in the motors



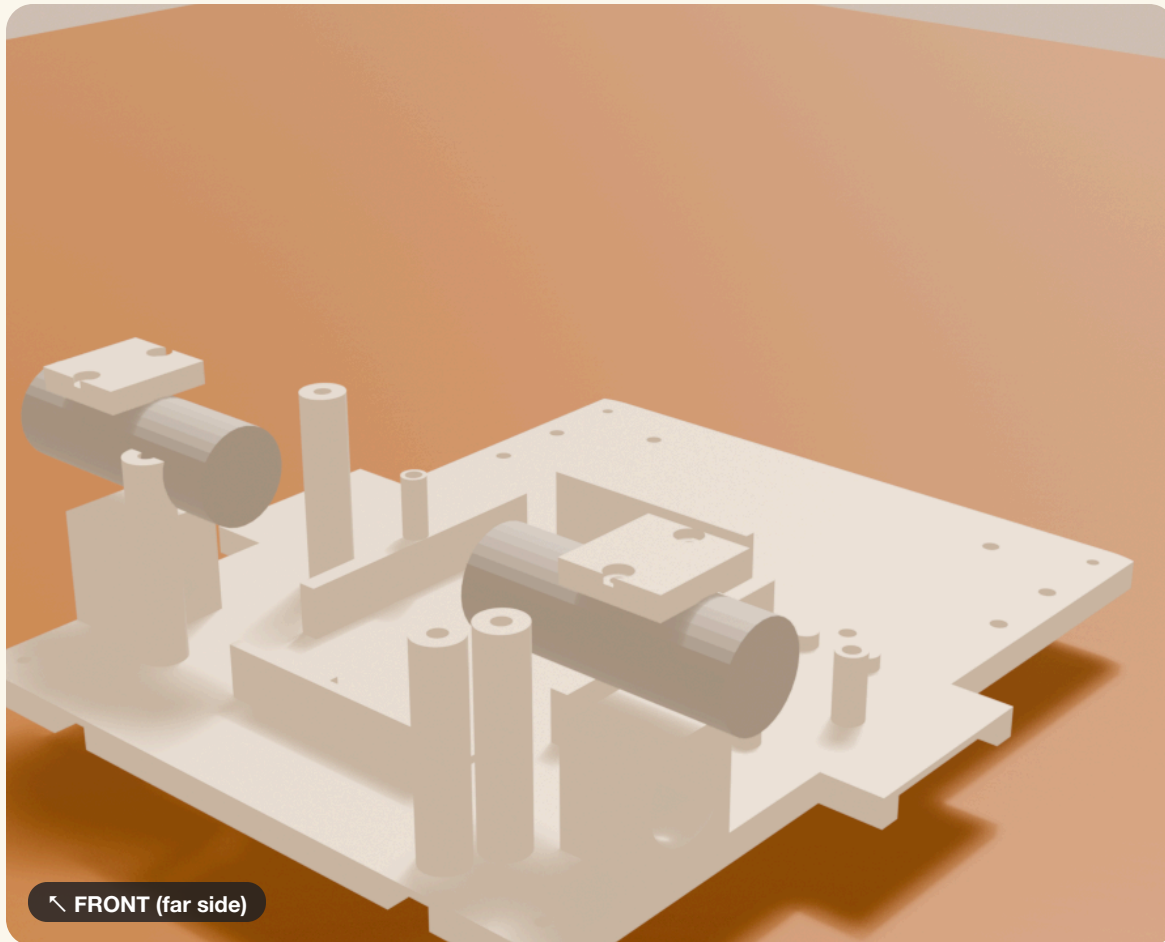
x2



x2



x4
M3 screws



1. Lay a motor in each cradle with the metal shaft poking OUT through the hole toward the wheel side.
2. Point the wires inward, toward the middle of the robot.
3. Set a printed cap over each motor and screw it down with 2 screws per cap - snug, not gorilla-tight.

CHECK Motors don't wiggle. Shafts spin freely when you twist them.

4

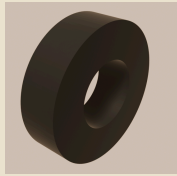
Make the wheels



x2



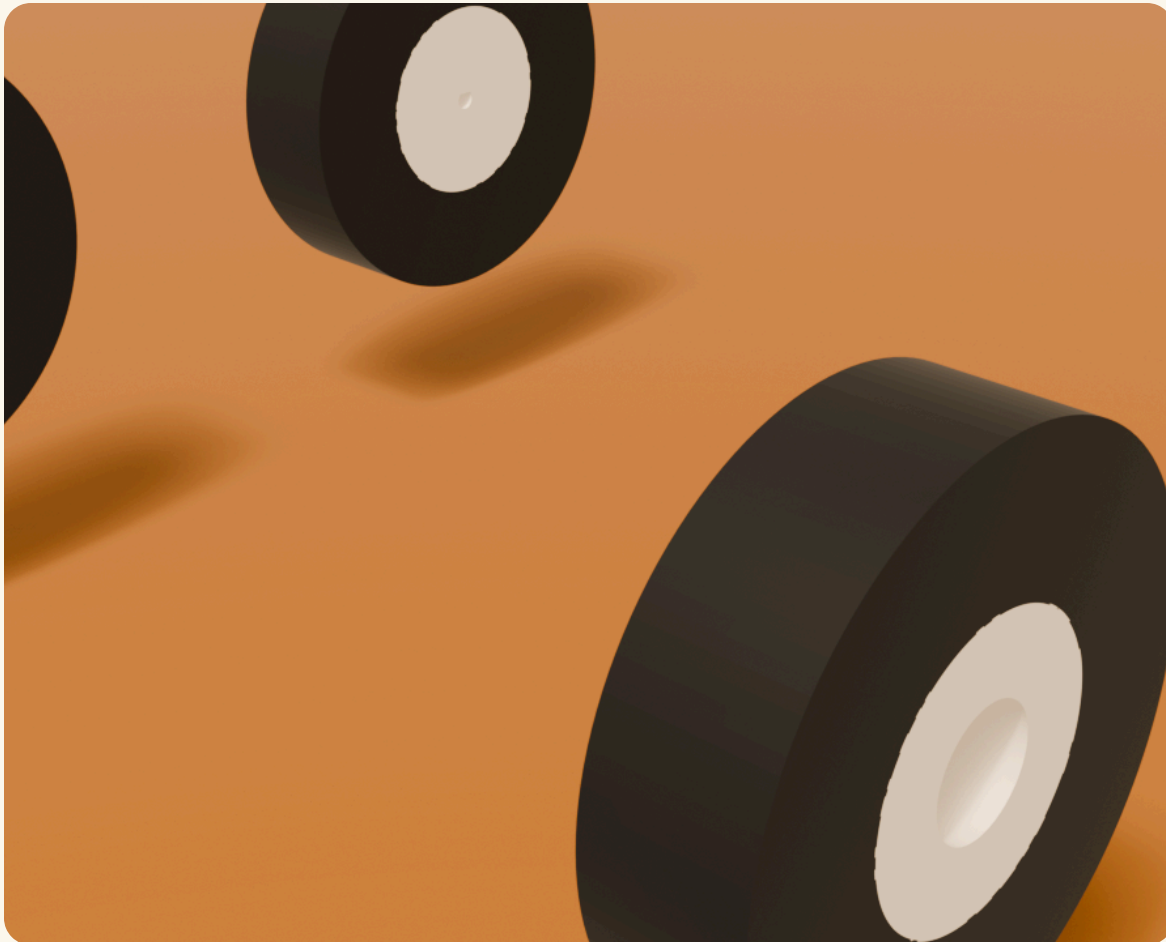
x2



x4



x2



1. GROWN-UP: melt 1 insert into the little bump on each BACK wheel's rim (2 total).
2. Stretch a stretchy printed tire over each of the four wheels, like putting a rubber band on a yo-yo.
3. Work it around evenly until it sits flat in the groove all the way around.

CHECK No tire bulges. All four look the same.

5

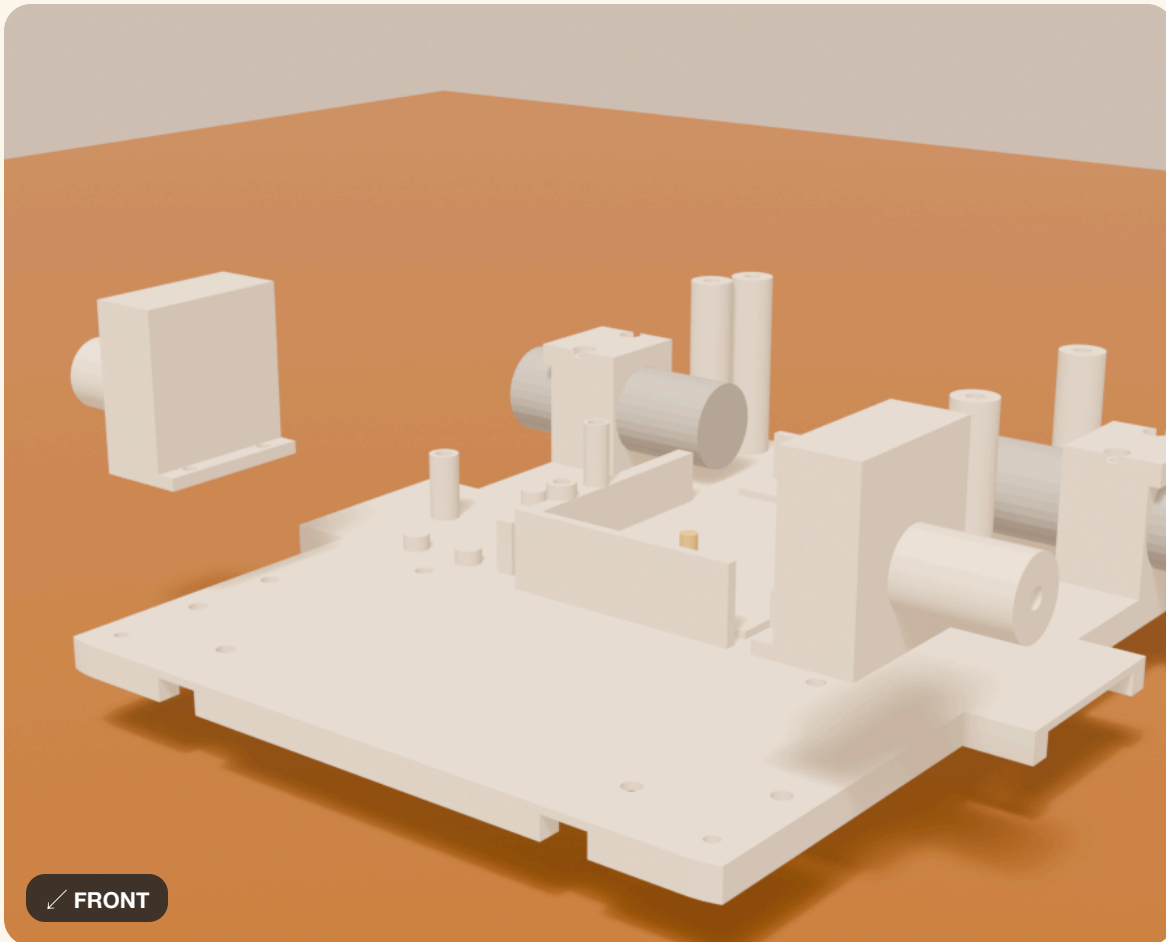
Bolt on the front pods



x2



x2

x4
M3 screws

1. GROWN-UP: melt 1 insert into the end of each pod's peg (2 total).
2. The two front pods have round pegs sticking out - those pegs are the front axles.
3. Screw each pod to the tray through its little foot tabs, 2 screws each, pegs pointing OUT.

CHECK Both pegs point straight out to the sides.

6

Put the wheels on



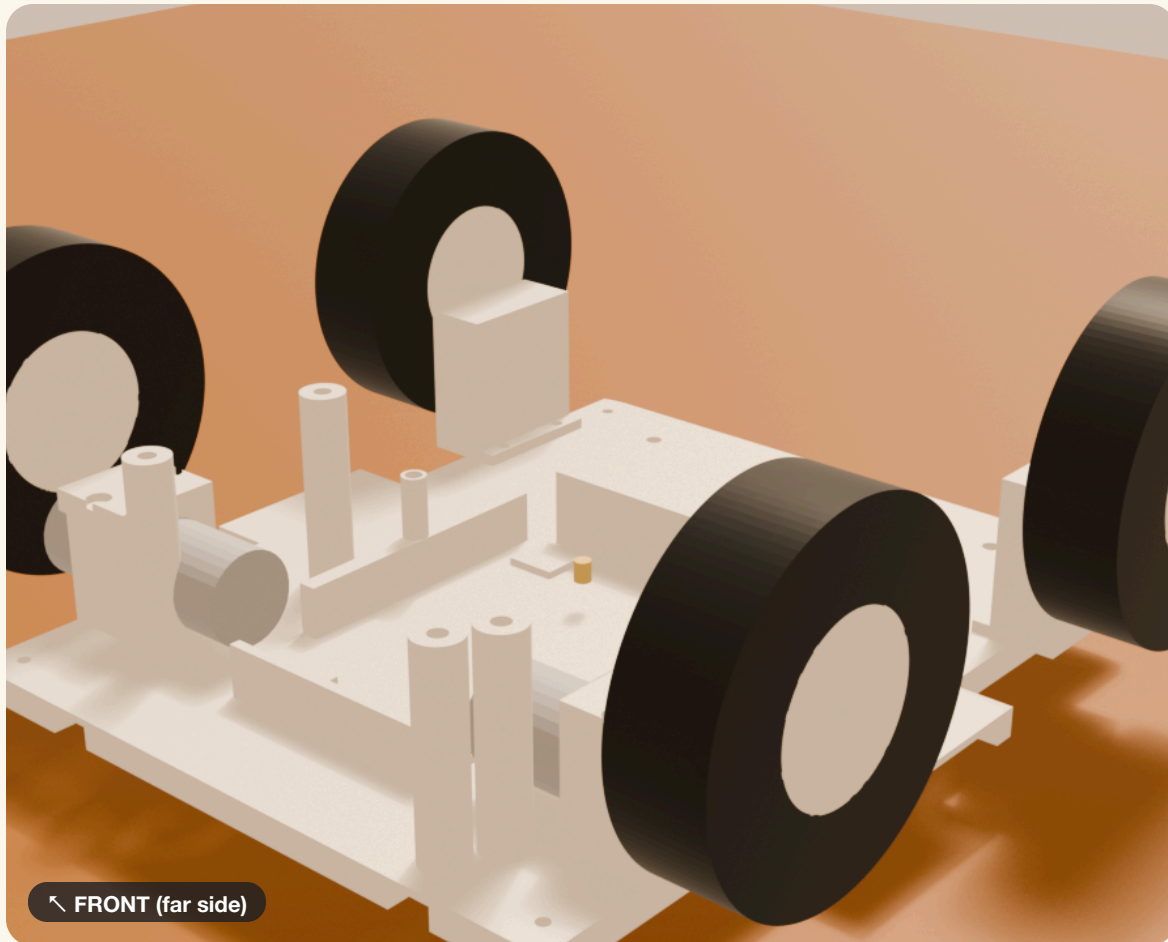
x2



x2



x2

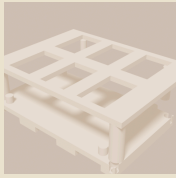
x4
M3 screws

1. BACK wheels: the hole has a flat side, and so does the motor shaft. Line the flats up, push the wheel on, then tighten the one clamp screw on the wheel's rim.
2. FRONT wheels: slide onto the pegs - they should spin freely. Put a printed washer on, then a screw into the end of the peg to keep the wheel from sliding off.
3. Don't overtighten the front screws: the wheels must still spin.
4. You printed extra washers - drop the leftovers in your spares box.

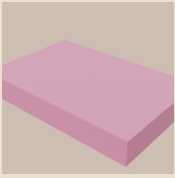
CHECK Back wheels should NOT spin freely by hand (the motor holds them). Front wheels spin freely.

7

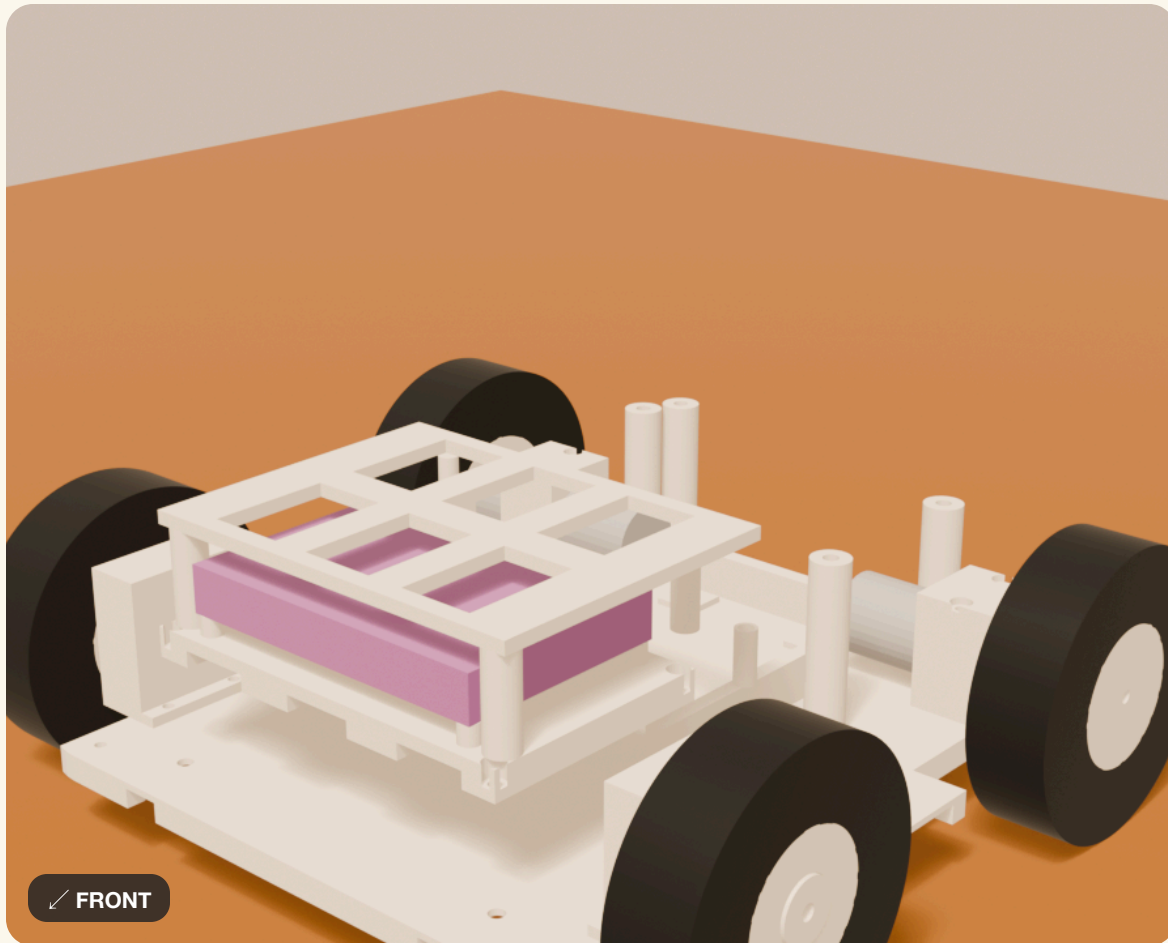
Build the brain tower (the controller tower)



x1



x1

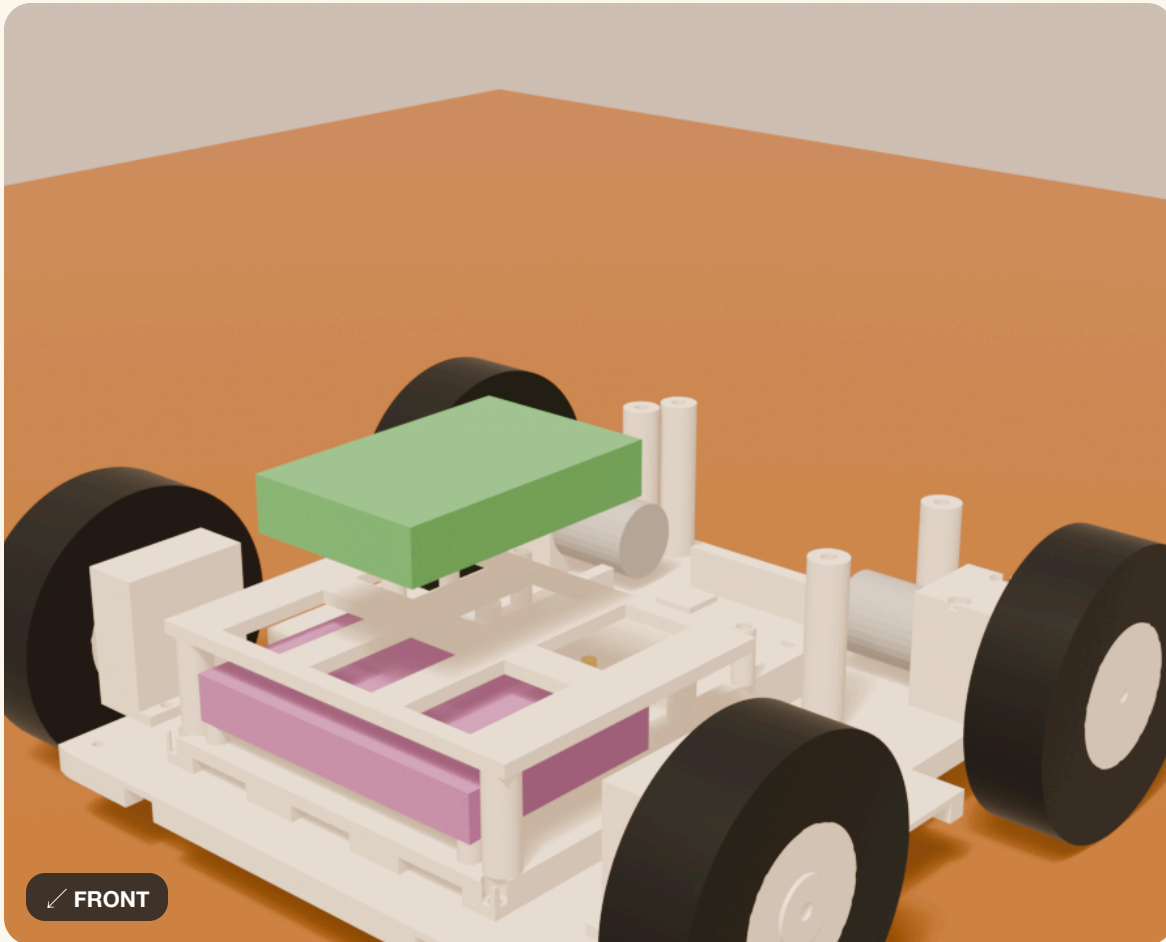
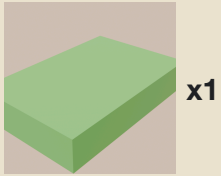
x4
M3 screws

1. Set the tower over the front-left of the tray - its screw holes match the four inserts.
2. Drive 4 screws down through the base tabs.
3. Rest the purple motor board on the four little posts, its green terminal blocks facing the LEFT side of the robot.

CHECK The board sits level on all four posts, terminals facing left.

8

Add the computer

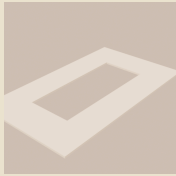


1. The Raspberry Pi lies flat on the tower's top shelf frame.
2. Its USB ports face the BACK of the robot so the cables can reach.
3. Don't screw anything - the shelf pocket holds it, and the head's cable will come down to it later.

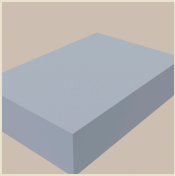
CHECK The Pi sits in its pocket, ports facing backward.

9

Strap in the battery



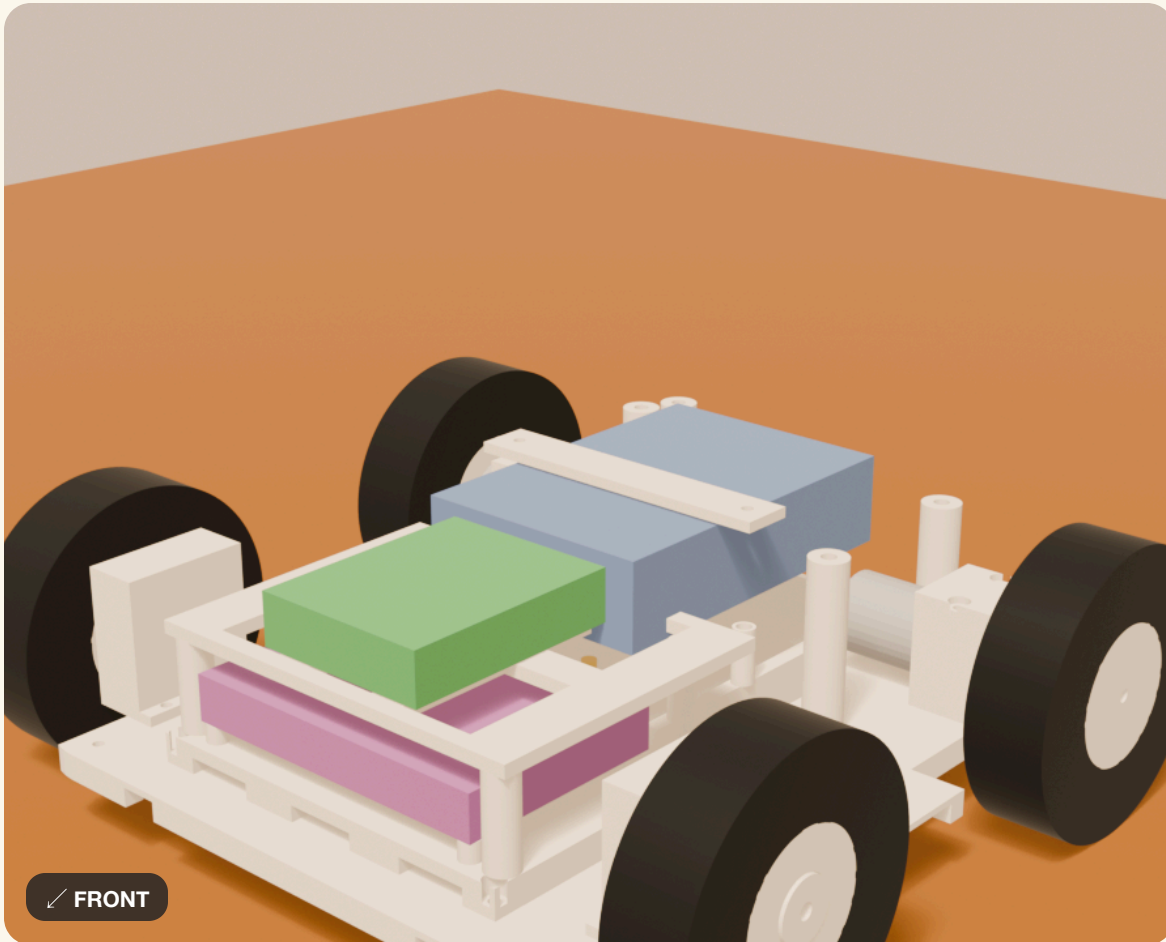
x1



x1



x1

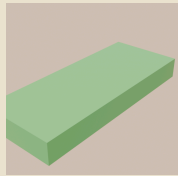
x2
M3 screws

1. Lay the soft TPU pad frame onto the four pads behind the tower.
2. Set the battery on it, wires pointing at the BACK of the robot.
3. Bridge the clamp bar across the battery onto the two posts and screw it down with 2 screws - firm, so the battery cannot slide.

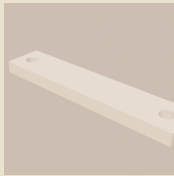
CHECK Grab the battery and try to wiggle it. It shouldn't move.

10

Add the safety helper



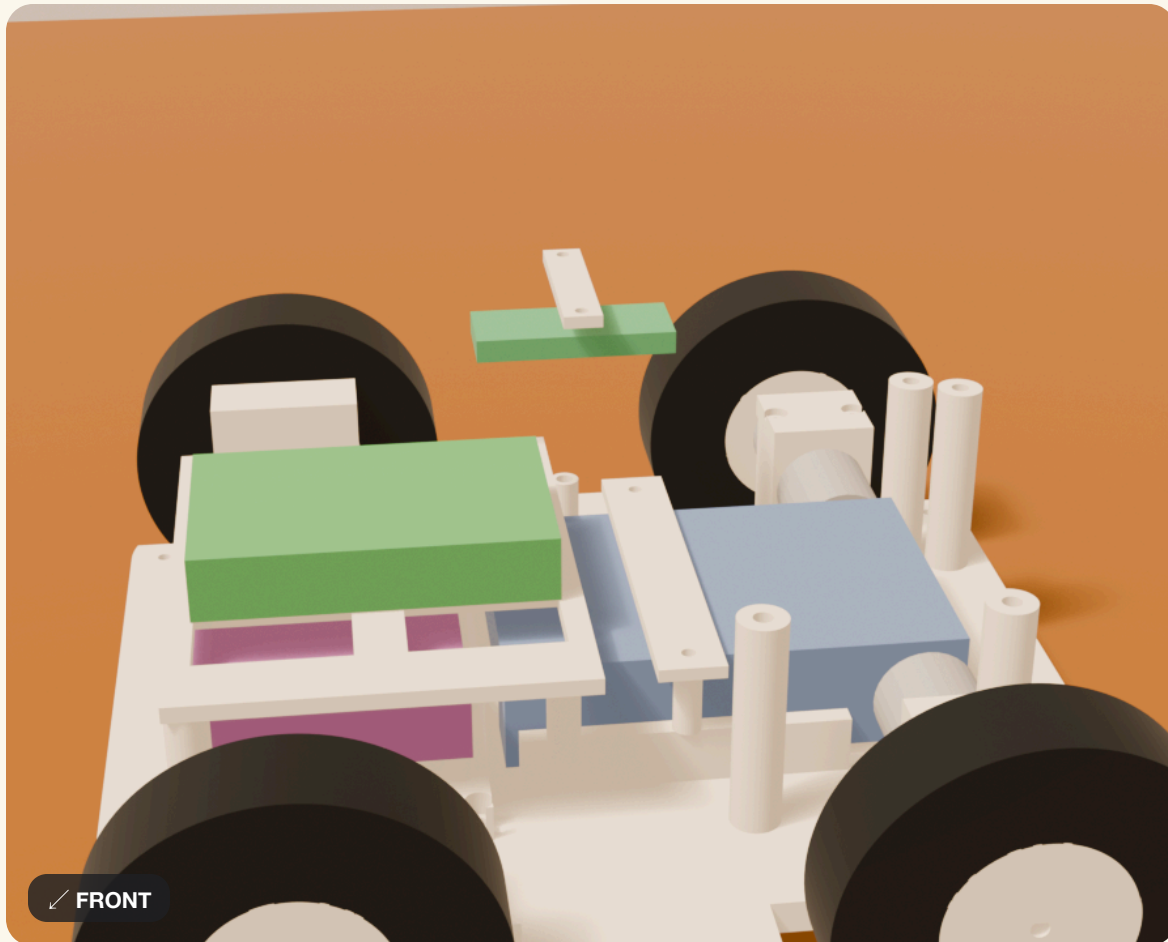
x1



x1



x2
M3 screws



1. The tiny green Pico sits on its little posts to the right of the tower, USB plug facing RIGHT.
2. Lay the small clamp bar across it and screw it down with 2 screws, gently - it's a small board.

CHECK The Pico is held snug and its USB port is reachable.

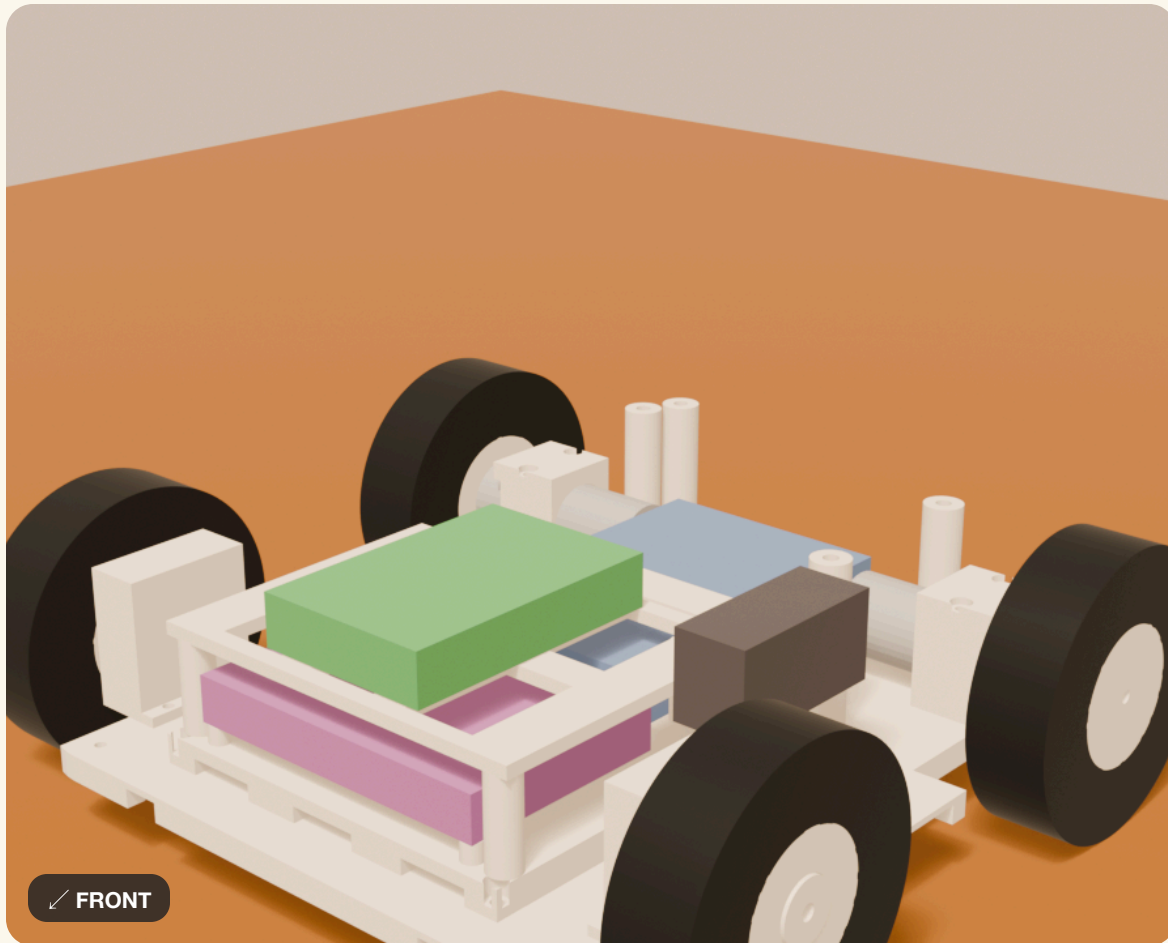
11

Mount the power relay

GROWN-UP STEP



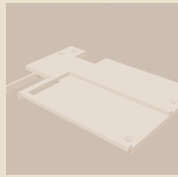
x1



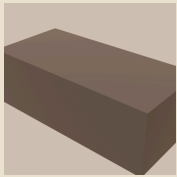
1. GROWN-UP: the relay drops into its floor pocket on the left, behind the tower.
2. Its metal bracket slots into the printed pocket; the terminals face UP so you can wire them later.
3. No wires yet - all wiring happens in the wiring chapter at the back of this book.

CHECK The relay clicks into its pocket and doesn't rattle.

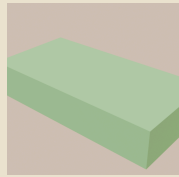
12 Put on the power deck



x1



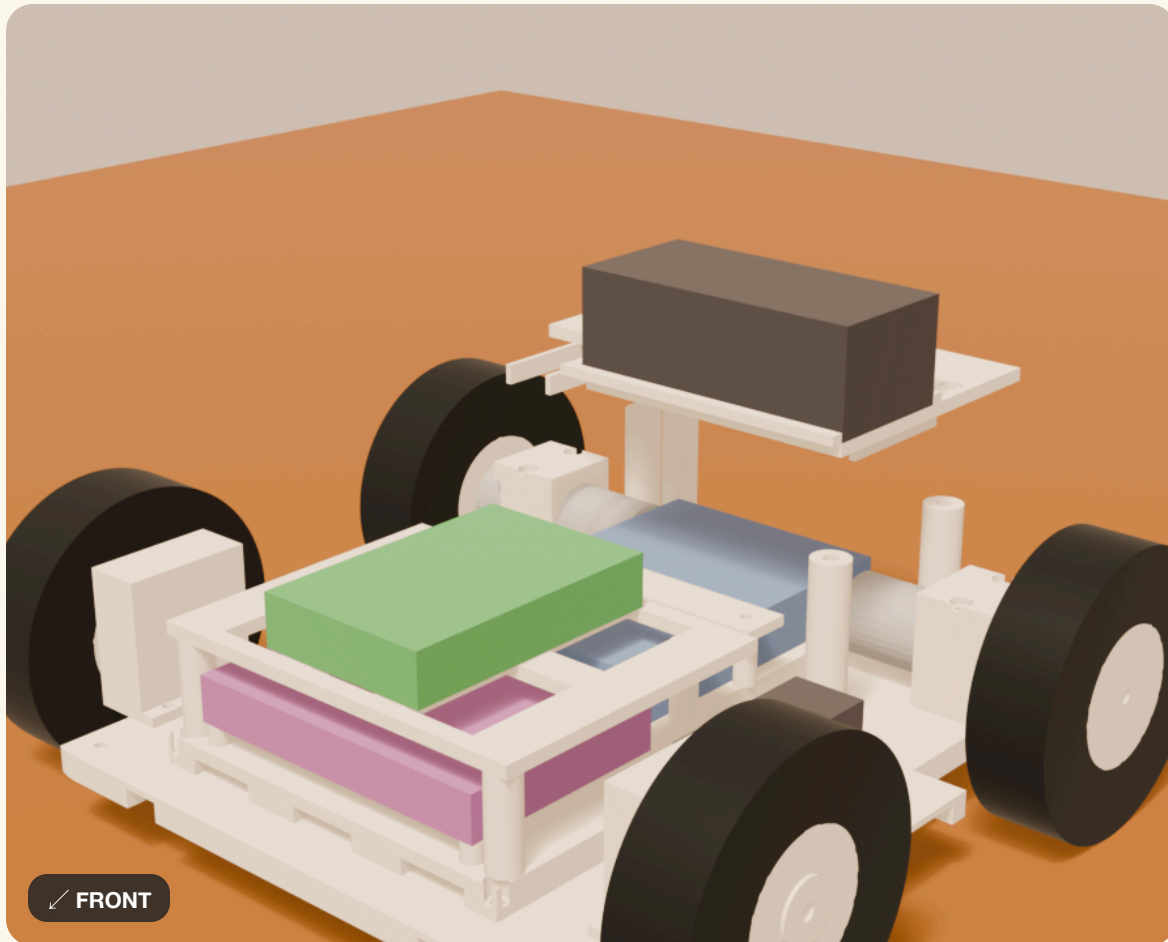
x1



x2



x4
M3 screws



1. First hang the two small green regulator boards under the deck's RIGHT end (they clip under; wires come later).
2. Lower the deck onto the four towers - the notch at the back-right corner goes around the battery wires.
3. Drive 4 screws down into the tower tops.
4. Set the black fuse box into its raised outline on the deck's right half, wire end hanging over the edge.

CHECK The deck is level and the fuse box sits inside its printed fence.

13 Lower the body shell

GROWN-UP STEP



x1



x14



x4

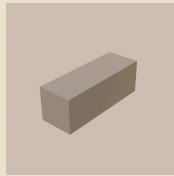
M3 screws



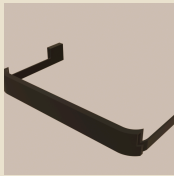
1. GROWN-UP FIRST: melt the shell's 14 inserts (4 corner lugs, 4 lid-ledge holes, 4 speaker posts, 2 sensor bosses).
2. Two people make this easy: lower the big shell straight down over EVERYTHING.
3. The wheel arches go around the wheels; the lip settles onto the tray edge.
4. Flip-check the underside: drive 4 screws UP through the tray's corner holes into the shell's lugs.

CHECK No gaps between shell and tray. The robot is now a box with wheels.

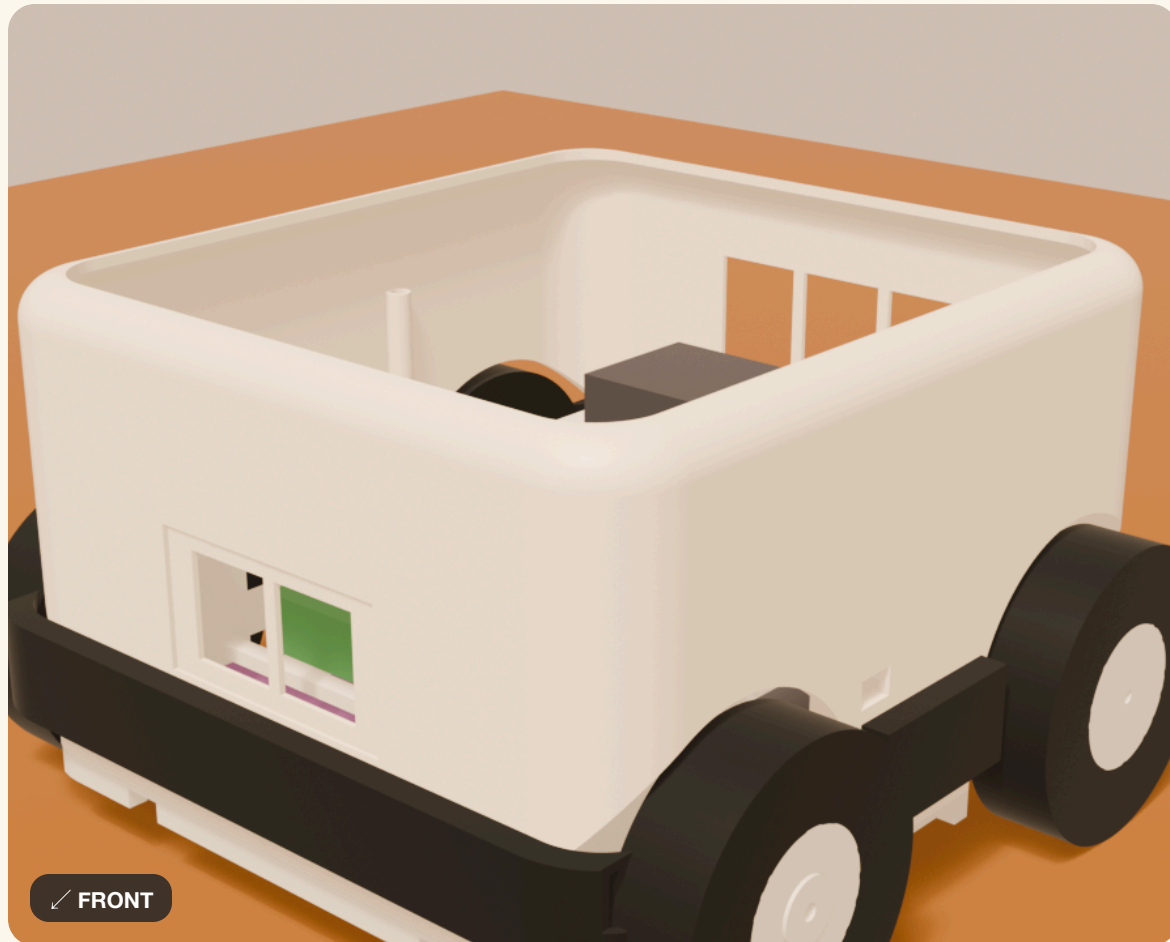
14 Feeler switches, then bumpers



x6



x2

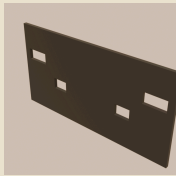


1. Tip the robot gently onto its side. Click each of the six little feeler switches into its pocket under the tray edge - two front, two back, one each side. Their tiny buttons face OUT.
2. Their wires tuck up inside for the wiring chapter.
3. Stand the robot back up. The two soft bumper halves wrap around the bottom, one from the front, one from the back.
4. They hug the body loosely on purpose - they squish in to press those feeler switches.

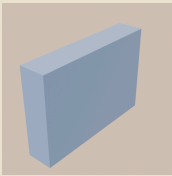
CHECK Press any bumper edge gently: it moves a tiny bit, you hear a soft click, and it springs back.

15

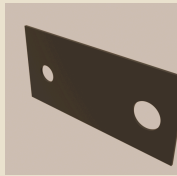
Face and back panels (with the front eyes)



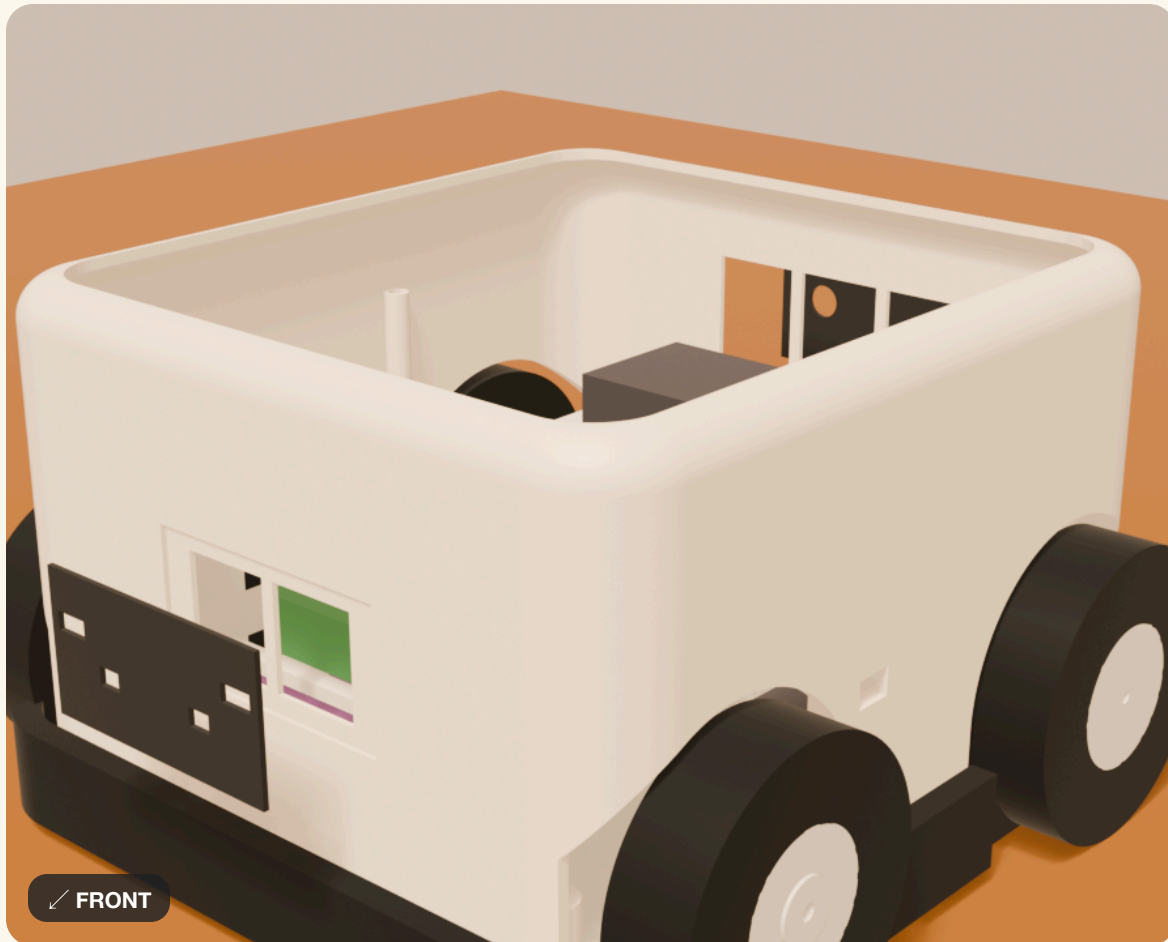
x1



x2



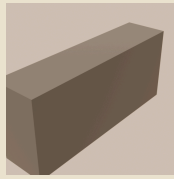
x1



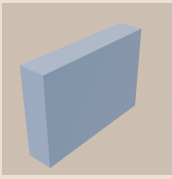
1. First press the two front distance boards into the pockets on the FACE panel's back - their little lenses peer through the two low holes.
2. Click the FACE panel into the front opening.
3. The dark BACK panel clicks into the back opening. Its two round holes are for the charger plug and the mute switch - the grown-up bolts those in with their own nuts during the wiring chapter.

CHECK Both panels sit flush; two tiny lenses look out of the face.

16 Speakers and distance eyes



x2



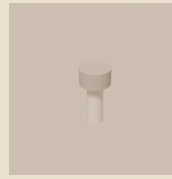
x2



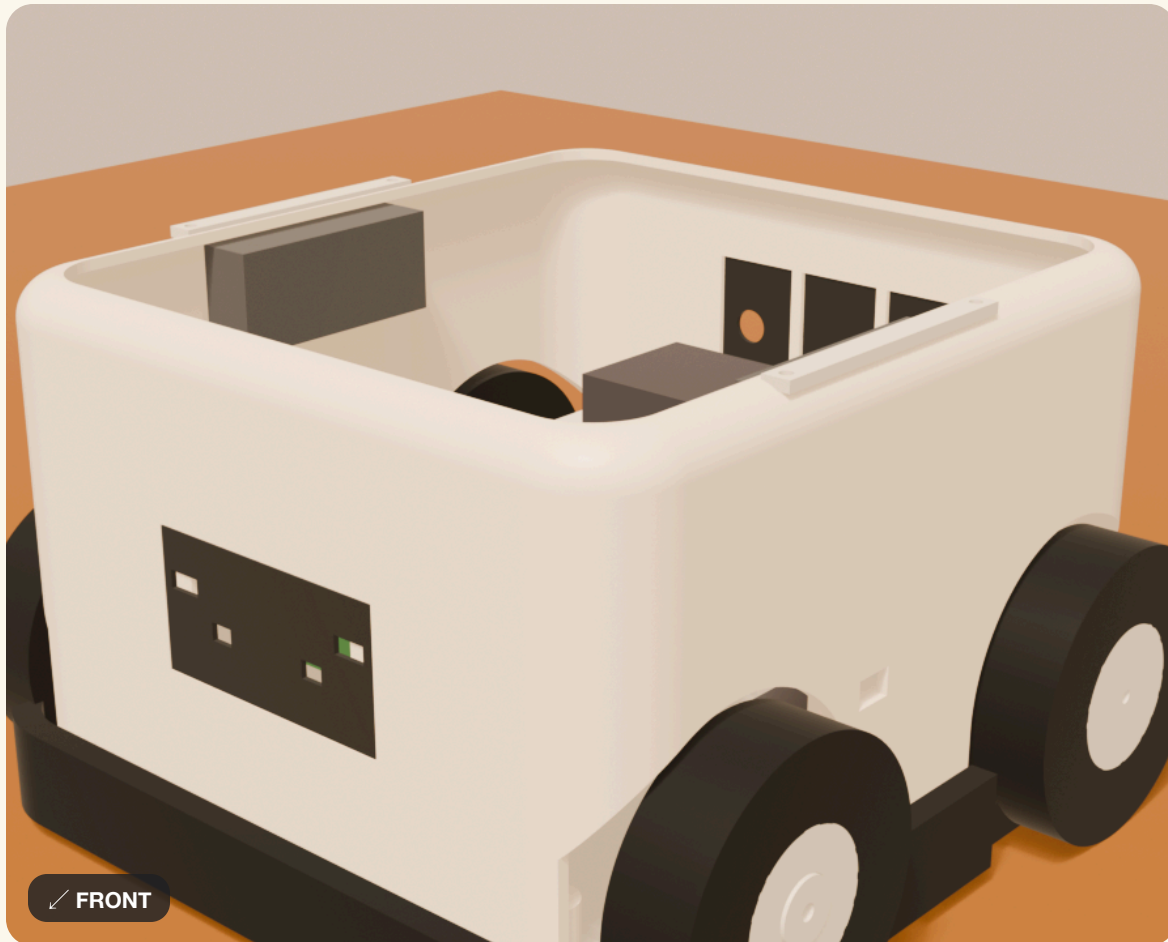
x2



x2



x6
M3 screws



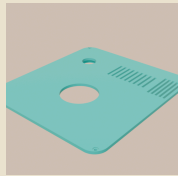
1. Reach in through the open top: rest a speaker on each side shelf, magnet side in, grille facing the wall.
2. Lay a clamp bar across each speaker's top and screw into the two posts (2 screws per side) - the inserts went in with the shell step.
3. Slide a little blue distance board behind each side window, then its clamp bar and 1 screw.

CHECK Speakers can't rattle; the blue boards peek through their side windows.

17

The lid and the BIG RED BUTTON

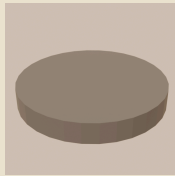
GROWN-UP STEP



x1



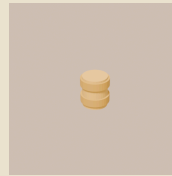
x1



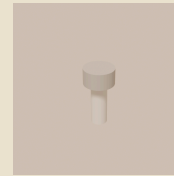
x1



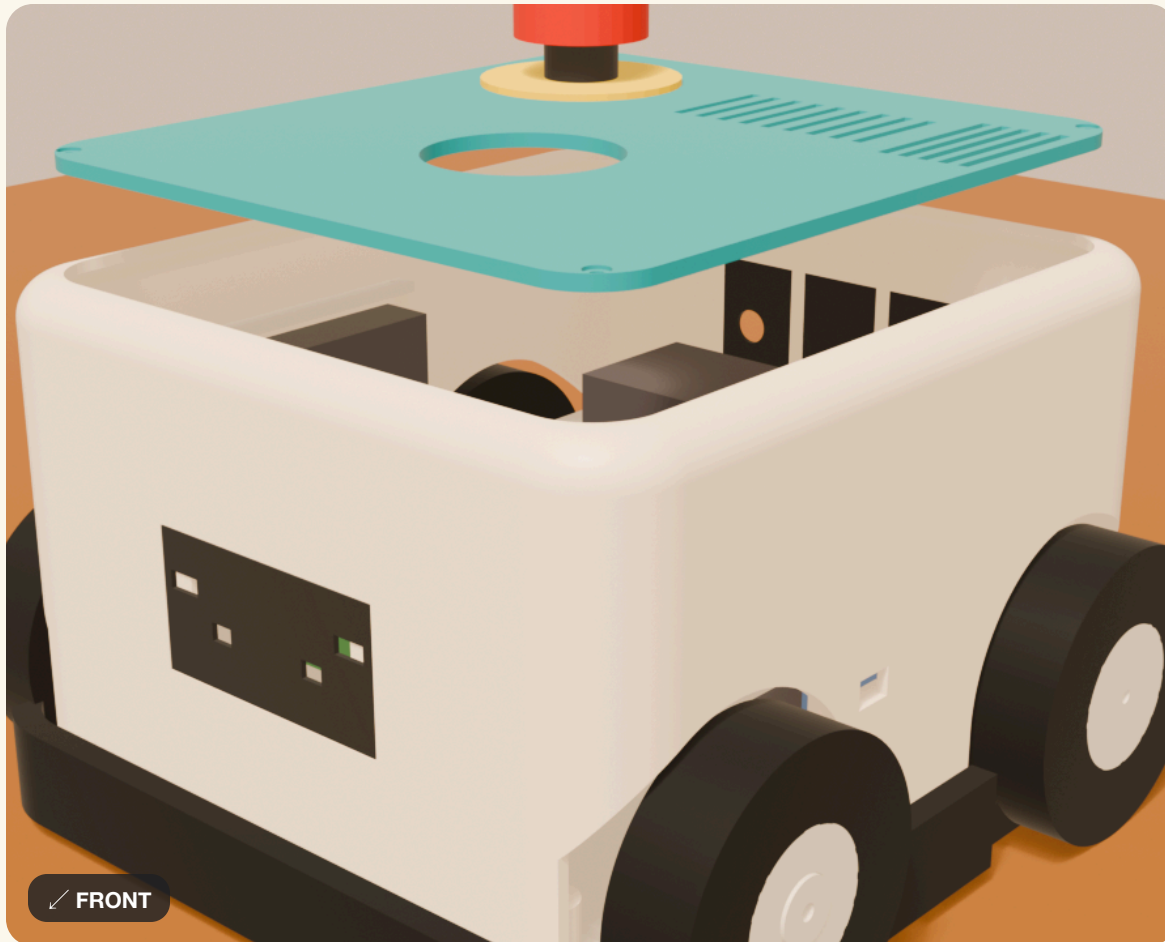
x1



x2



x2
M3 screws



1. GROWN-UP: melt the lid's 2 mic-boss inserts, then drop the red emergency-stop through the lid's round hole and spin its nut on underneath - the lid IS its mounting panel, and the printed ring under the lid makes it strong.
2. Set the round microphone under the lid's slotted area and screw its ring cradle to the two bosses (2 screws).
3. Rest the lid in its ledge - DON'T screw it yet. The neck, the head, and all the grown-up wiring still need the inside. Its 4 corner screws are the very last thing in this book.

CHECK Slap test: the red button clicks down hard and twists to release.

18

Grow the neck



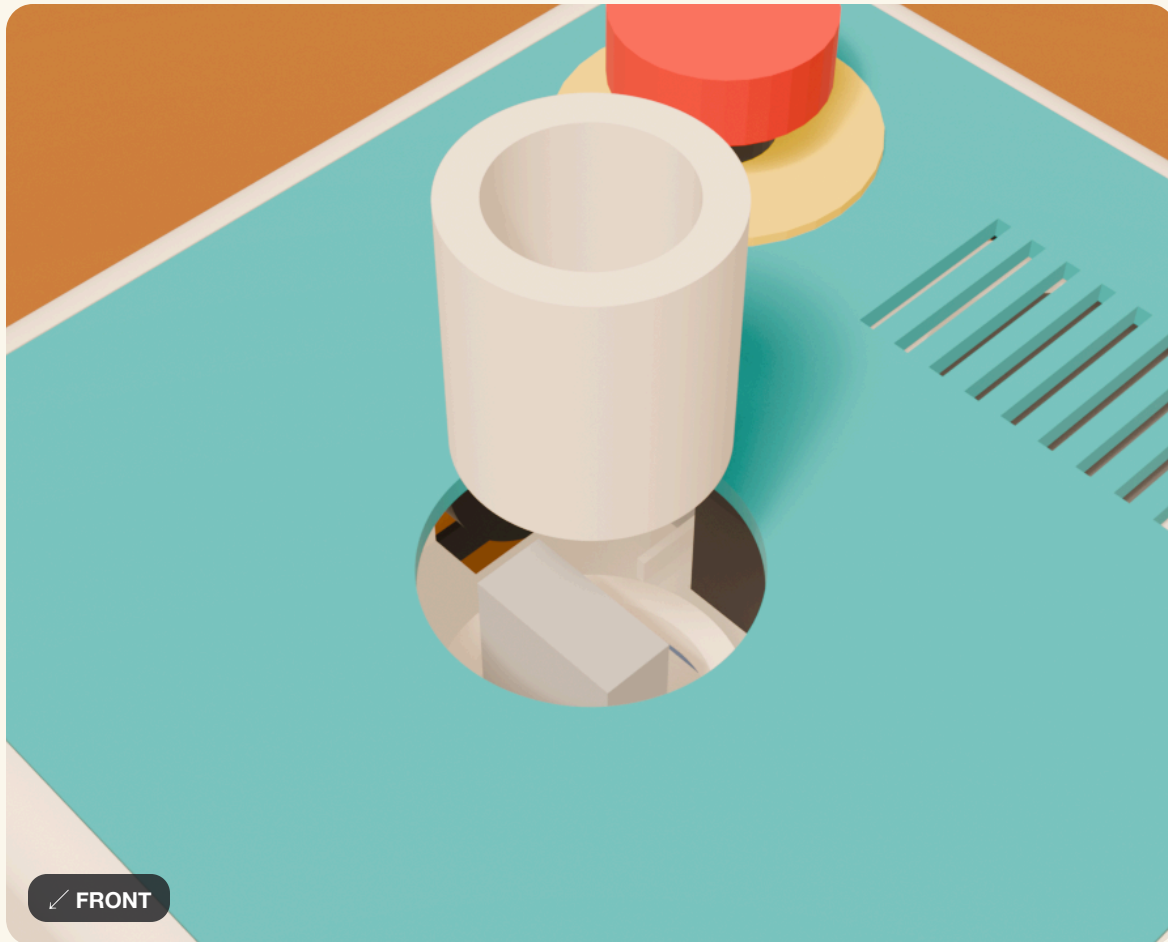
x1



x1



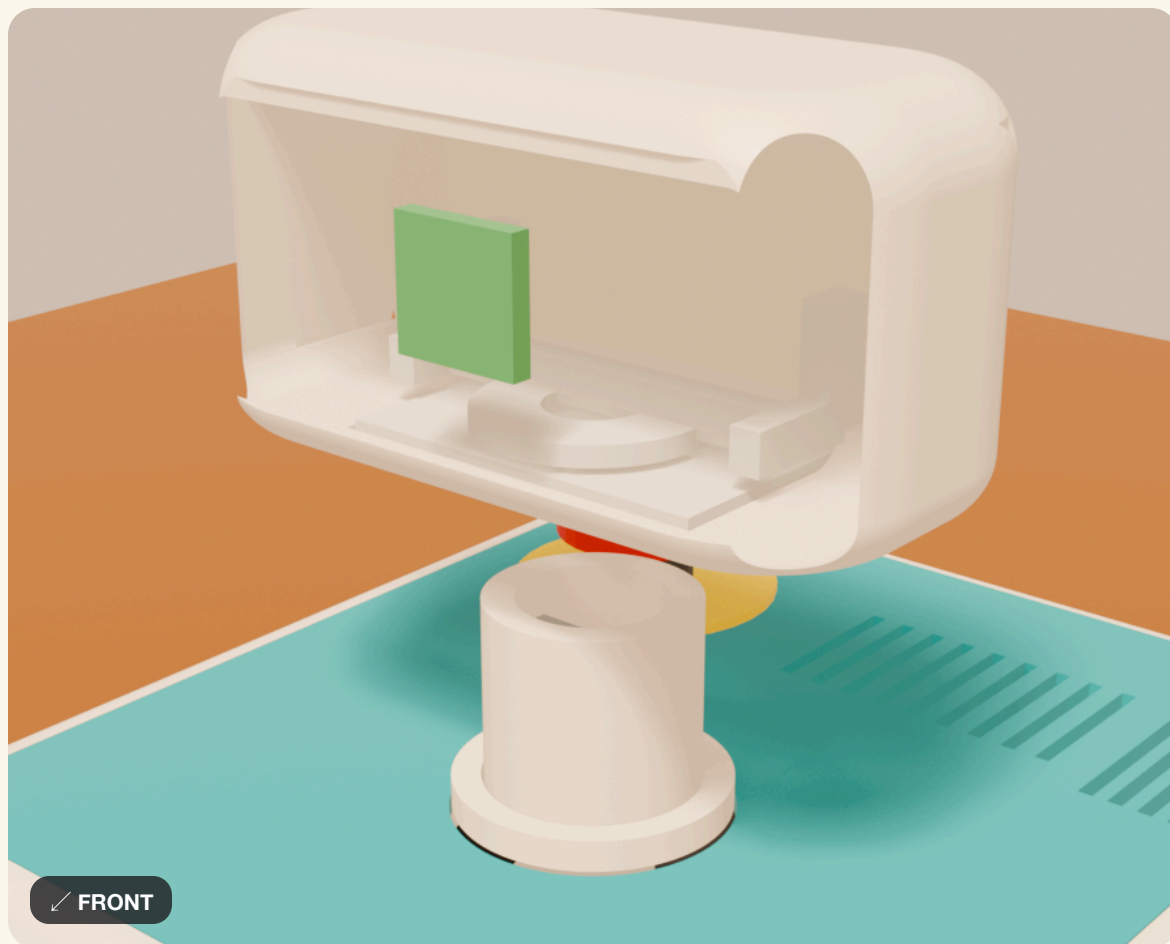
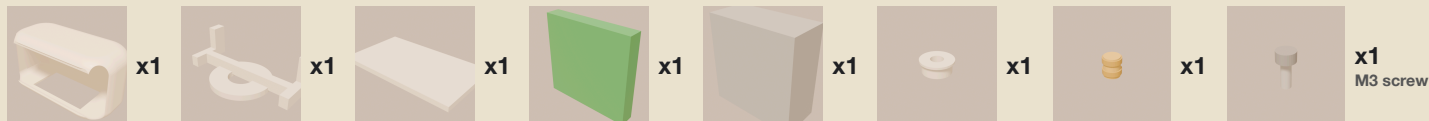
x1



1. Feed the neck tube down through the lid's front hole.
2. From inside, twist the bayonet collar onto the neck's bottom - a quarter turn locks it, like a camera lens.
3. The neck servo sits in the collar's cradle underneath (its wire joins the wiring chapter).

CHECK The neck turns smoothly by hand and cannot pull up and out.

19 Build the head

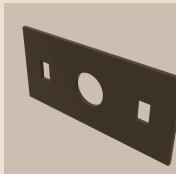


1. Grown-up melts the last insert (the tilt-pivot side), then: screw the yoke's ring onto the top of the neck.
2. Slide the camera into the pocket behind the face opening, lens forward; its flat ribbon cable runs down through the hollow neck to the Pi.
3. Lower the head shell over the yoke: one side takes the tilt servo, the other side gets the little bushing and 1 pivot screw.
4. Close the underside with the pan plate.

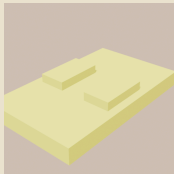
CHECK The head nods up-down and the neck turns left-right, smoothly.

20

Give it a face



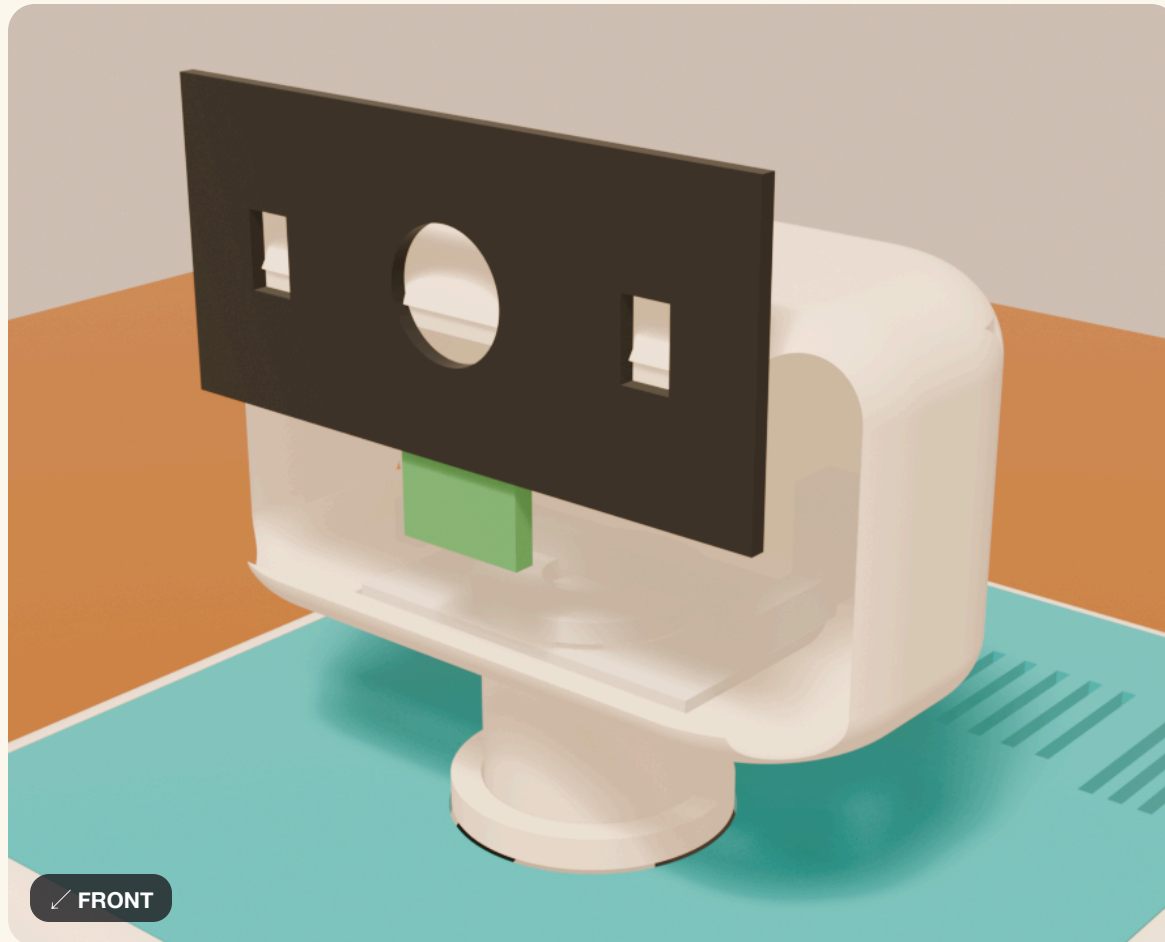
x1



x1



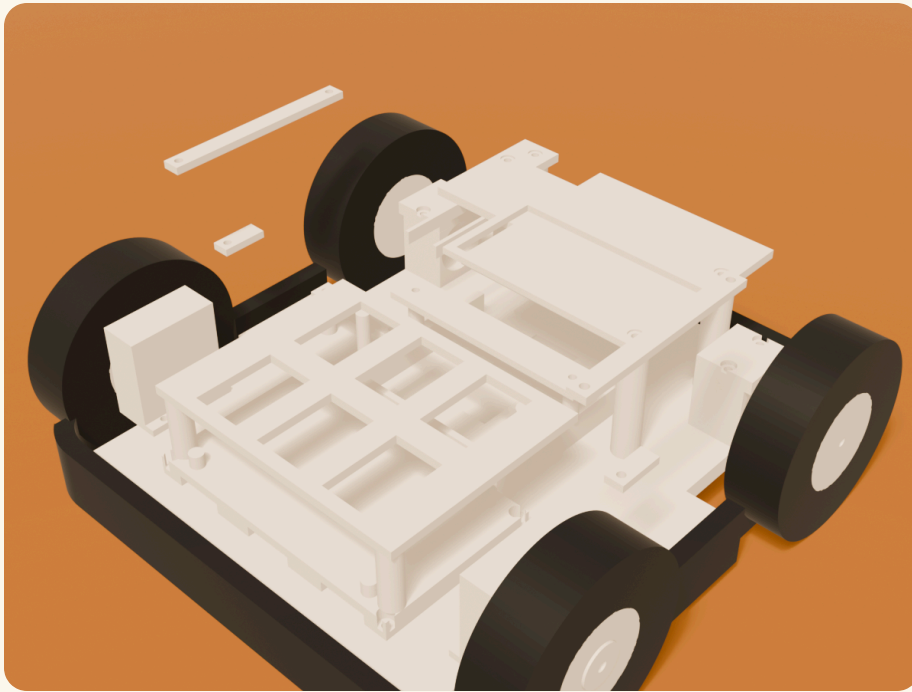
x1



1. Clip the lime eye bar behind the face panel's eye slots (glow boards ride behind it).
2. Press the face panel into its recess: camera hole over the lens.
3. The second lime bar clips behind the front body panel's light slots - each glow board rides in the pocket behind its bar.
4. Leave the 4 glow boards in their bag for now - the grown-up seats them in those pockets during the wiring chapter.

CHECK Rover Bean is looking at you. Say hi.

Chapter 4 - Where every electronic part lives



Part	Home	Faces
MDDS10 motor board	tower posts, front-left	terminals LEFT
Raspberry Pi 5	tower top shelf	USB ports BACK
Pico 2	floor, right of tower	USB RIGHT, pins UP
Battery	middle-back floor	wires BACK
Relay	floor pocket, left	terminals UP
Fuse block	deck, right half	wire exit RIGHT edge
Regulators (5 V + 6 V)	hang UNDER deck, right end	6 V wire exit LEFT
Big red button	through the lid	UP (slap it!)
Mic array	under the lid slots	UP
Speakers	side shelves	grilles OUT
ToF distance boards	2 behind face, 2 side windows	lenses OUT
Bumper feeler switches x6	pockets under the tray edge	buttons OUT
Charger inlet + mute	back panel holes	BACK
Camera	head, pocket behind the face opening	lens FRONT
Pan + tilt servos	pan inside the neck, tilt in the head's side	pan shaft UP, tilt shaft SIDEWAYS
Glow boards x4	pockets behind the eye + status bars	lights OUT

The two speaker amp boards (MAX98357A) zip-tie beside their speakers - a future version of the body adds printed pockets for them.

Chapter 4 - Grown-up wiring rules

This whole chapter is grown-up work. The builder can watch and hand you zip ties.

The battery stays OUT of the robot until every wire is checked against these maps.

Use wire colors: RED = battery 12 V, YELLOW = switched motor 12 V, BLUE = 5 V, GREEN = 6 V, BLACK = ground, WHITE = signals.

Crimp or solder every joint; no bare twists. Label both ends of every wire with tape.

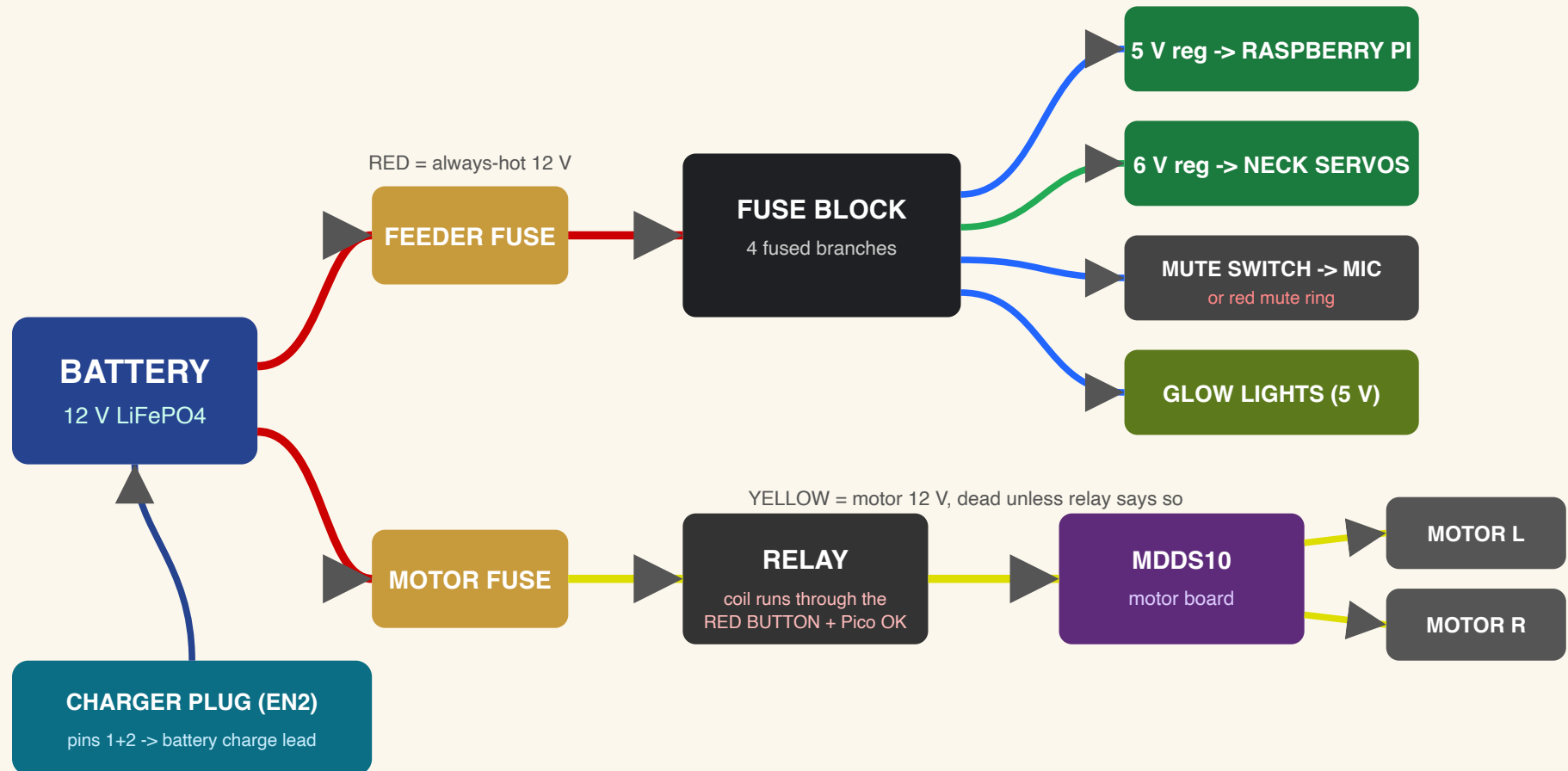
Fuses go in LAST, after a meter check - the sizes are on the shopping page (2-3 A branches, 5 A motor, 7.5 A feeder).

The robot must FAIL STOPPED: if any of this feels wrong, it stays off.

The power map (every 12 V wire)

From	Through	To
Battery +12 V	feeder fuse (near battery)	Blue Sea fuse block IN
Fuse branch 1	D24V90F5 regulator	5 V to the Raspberry Pi
Fuse branch 2	D36V50F6 regulator	6 V to both neck servos
Fuse branch 3	mute switch common	mic USB power OR red mute ring
Fuse branch 4	5 V accessories	NeoPixel eyes + status lights
Battery +12 V	motor fuse -> relay contacts	MDDS10 motor board power
Relay coil 12 V	through BOTH red-button NC contacts	coil ground via Pico enable
Charger EN2 pins 1+2	direct to battery charge lead	pin 3 = charger-present to Pico

Chapter 4 - Power, as a picture



Chapter 4 - The signal map (thin wires)

From	To	What travels
Raspberry Pi	MDDS10	serial motor commands (through the Pico's watchful eye)
Raspberry Pi	Pico 2	USB: heartbeat + status; Pico can always stop motors
Motor encoders (6 wires each)	Pico 2	wheel speed feedback
Bumper switches x6	Pico 2	any press = stop, instantly, no software needed
ToF boards x4	Raspberry Pi	I2C daisy chain (STEMMA cables)
NeoPixels x4	Raspberry Pi	one data line, chained eye->eye->status->status
Camera	Raspberry Pi	flat FPC ribbon down the hollow neck
Servos x2	Raspberry Pi	PWM signal wires (power from the 6 V rail)
Mic array	Raspberry Pi	USB (its 5 V passes through the mute switch)
Pi I2S pins	MAX98357A amps	digital sound out; the amps zip-tie beside each speaker (a future version adds printed pockets)

Cable neatness

Quiet wires (signals, audio, sensors) ride the LEFT channel under the deck. Power wires ride the RIGHT channel. The camera ribbon goes up the little tunnel beside the neck servo, through the hollow neck, into the head. Zip-tie at every printed tie point; no wire may touch a wheel or the fan.

Before the battery goes in - the grown-up meter checklist

No shorts first: beep-test + to - at the fuse block (no fuses in yet). Then:

1. Red button held down = relay coil circuit reads OPEN.
2. Regulator outputs read 5.0-5.2 V and 6.0 V on the bench before their loads connect.
3. Fuses go in smallest-first, one branch at a time (the sizes are on the shopping page).
4. Charger plug in = motors will not run.
5. Only then: drive the lid's 4 corner screws - the last 4 of the robot's 41. NOW it's closed for good.

You built a robot!

High five, builder. Rover Bean's body is done.

Chapter 5 - Before it ever moves (grown-ups)

Slap the red button while the wheels spin on blocks - everything must stop instantly. Press each bumper - stop. Unplug the Pi's heartbeat - stop. Plug in the charger - it refuses to drive. Only after every one of these passes does Rover Bean get floor time, supervised, at walking pace.

Load the brain (grown-up)

Flash **Raspberry Pi OS** onto the microSD with Raspberry Pi Imager, then follow the README at github.com/brianpattison/robot to install the robot's programs on the Pi and the safety code on the Pico. The tests above only work once the brain is loaded.

The robot's number one rule, forever: **when anything is wrong, it stops.**

Codex Rover Bean v2 - printed-only body - one screw, one key, zero supports.

